Molecule identification from MS/MS for CASMI 2026: one change per submission, from 0.115 to 0.456

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Abstract

We rank candidate structures for 400 molecules from their tandem mass spectra in the CASMI 2026 Kaggle competition, scored by MRR@25 on the InChIKey connectivity block. The first submission scored 0.115 against a local estimate of 0.664; the estimate came from labels drawn from the same data as the candidates, and was retracted. Since then every number reported is a public leaderboard score, and each of the 18 submissions changed one thing (a nineteenth re-ran v19 unchanged). A PubChem-scale pool with a popularity prior, a learned spectrum-to-fingerprint model, and two forward models that predict each candidate's spectrum take the score to 0.349; a second PubChem pass that avoids structures similar to the copied train rows takes it to 0.356, against 0.421 for the public notebook these components come from. Running that notebook unchanged as our own kernel scored 0.415, +0.059 over the best of our own pipeline. Running a later public notebook, E006 with PubChem promotion at *rel* > 150 (public 0.446), unchanged as our kernel reproduced its public score exactly: 0.446, +0.031 over the first fork and +0.090 over our own best. It was the base every later change was measured against: raising its engine-2 fusion weight from 0.6 to 0.8 scored 0.437, and a third public notebook (0.451 upstream at its version 52) forked unchanged at its current version 53 scored 0.445. A fourth public notebook of the same lineage, which adds PubChemLite structures to the second engine's library (public 0.456), reproduced its score exactly as our kernel: 0.456, +0.010 over the E006 fork, and the current best. A de novo route was closed offline before any submission: FRIGID, given the true formula, placed the answer in its top 10 for 0 of 50 held-out molecules against a pre-registered bar of 5. 25 submissions were made and all were scored. No local estimate of the score exists.

1 Task

Each test molecule has one or more LC-MS/MS spectra (400 molecules, 1213 spectra). A submission lists up to 25 SMILES per molecule; the score is the mean reciprocal rank of the answer, matched on the first 14 characters of the InChIKey. The competition runs notebooks offline, so every database and model is attached as a dataset. Only the public split of the leaderboard is visible.

2 Method

Current best: a fork of the public 0.456 notebook (pubchemlite-fork). The submission is Hongbin Gu's public notebook *casmi26-v465-v15-pop015*, run byte-identical (md5 `9d1ab164...`) as the private kernel `charyeezy/casmi-pubchemlite-fork`, version 1, on the same GPU image as the other forks, with 15 public datasets attached. It is the E006 pipeline described next, through the 0.451 recipe (promotion at *rel* > 150, the rank-2 hedge, standardised ChEBI/LIPID MAPS candidates), with three changes: PubChemLite structures added to engine 2's library (0.455 upstream), GLACIER proxy scores from other adducts for molecules without an $[M+H]^+$ spectrum (N1A), and the PubChem popularity weight at 0.15 instead of 0.25 (0.456).

Previous best: a fork of the public E006 notebook (e006-fork, 0.446). The submission was the public notebook *CASMI26 | E006 + PubChem Promo rel>150 | LB 0.446*, run byte-identical (md5 `9829a966...`) as the private kernel `charyeezy/casmi-e006-fork`, version 2, on a T4. Its lineage is flexonafft's v80 recipe (public 0.436), then E006 (0.442), then the PubChem promotion's structure-evidence clause at $rel > 150$ instead of 250 (0.446). Three candidate sources run side by side: engine 1 is the v4n pipeline of the previous fork (below, top 60); engine 2 is a separate two-ranker engine over COCONUT, ChEBI/LIPID MAPS and train structures, with simulated ranker rows (top 40); and a PubChem-only channel ranks up to 25 non-pool structures by $z(f \cdot z) + 0.25\times$ popularity. ICEBERG and GLACIER (GLACIER on $[\mathrm{M+H}]^+$ only) score the same-formula groups of all three lists. PubChem structures are merged into the engine-1 list when the library's maximum similarity is below 0.9, and the first is promoted to rank 1 when $(S > 6 \wedge pop \geq 5) \vee rel > 150$, where *rel* is how far the best PubChem $f \cdot z$ exceeds the best pool candidate's. The merged list is fused with engine 2 by weighted reciprocal rank, $1/(3 + r_1) + 0.6/(3 + r_2)$, re-ranked by GLACIER within formula groups, deduplicated by the official metric identity with one vote per identity per engine, re-ranked by ICEBERG where GLACIER has no scores, and finally engine 2's top structure is inserted at rank 2 when it differs from rank 1 (`rank2`).

First fork: the 0.421 notebook (fork-v1, 0.415). From the first fork on 17 October the submission was the public notebook *CASMI26 | v4n Fusion + PubChem on public 0.421*, run unchanged as the private kernel `charyeezy/casmi-v4n-fork` on a T4. It searches every test spectrum against all of train and propagates matches to analogs; ranks a pool of train ∪ COCONUT ∪ generated derivatives with a public LightGBM ranker (`ranker_v4b.pkl`, 160 features, 4 boosters); joins PubChem candidates only when the library's maximum similarity is below 0.7; and re-scores with ICEBERG and GLACIER (off for library hits), falling back to fragment rescoring where they cannot run. Where our pipeline excluded the train structure behind each copied spectrum, the fork uses train matches as evidence.

Our own pipeline (best 0.356, v19), described below, is the record of what we built before the forks.

1. **Candidates.** Within a 5 ppm window of the molecule's neutral mass, the 1000 best-documented PubChem connectivities (by $\log(1 + \text{substances}) + \log(1 + \text{PubMed})$) from a 106-million-structure tier, plus train structures in the window.

2. **Copied spectra.** 1189 of 1213 test spectra are bit-identical to a train row; every molecule has at least one. The train structure behind it is removed from the candidates.

3. **Ranking.** An ensemble of two public FPNet checkpoints predicts fingerprint logits z per spectrum and per polarity-merged spectrum; a candidate with fingerprint f scores $f \cdot z$, z-scored within the molecule, plus $0.25\times$ popularity.

4. **Forward models.** For the top 60, ICEBERG and GLACIER predict every candidate's spectrum. Within each same-formula group reaching the top 25 the order becomes $z(\text{score}) + \lambda\, z(\text{ICEBERG}) + \lambda\, z(\text{GLACIER})$, with $\lambda = 1$.

3 The funnel

stage	surviving
test molecules	400
candidates per molecule (median, max)	1038, 1445
copied train structures removed	400 (one per molecule)
molecules with $[M+H]^+$ / $[M+Na]^+$ spectra (forward-model coverage)	371
candidates scored by each forward model	21,865 of 23,552 sent
molecules whose top-1 the forward models change ($\lambda = 1$)	208

4 Results

Every score is the public leaderboard's. Twenty-five submissions were made, including one identical re-run; all were scored; none is omitted. One further run, the E006 fork's kernel version 1, failed its own GPU preflight and was not submitted (below).

v	change from the previous kept version	public MRR@25
1	fragment explainability over train's 275,798 structures	0.115
4	+ PubChem tier, 4000 nearest-mass per molecule	0.049
5	popularity shortlist of 300, +0.25× popularity	0.135
6	FPNet in place of fragment explainability	0.250
7	FPNet ensemble (two checkpoints)	0.258
8	popularity weight 0.15 (not kept)	0.254
9	shortlist 1000	0.261
11	copied train structures removed	0.262
12	ICEBERG, CPU, 269 of 371 molecules scored	0.286
13	ICEBERG on a T4, all 371	0.292
14	+ GLACIER, $\lambda = 0.5$	0.320
15	$\lambda = 1.0$	0.349
16	GLACIER on $[M+H]^+$ only (not kept)	0.345
17	$\lambda = 1.5$ (not kept)	0.346
18	+ spectrum-driven second PubChem pass (not kept)	0.353
19	second pass skips Tanimoto ≥ 0.5 to the copied structure	0.356
19	identical re-run	0.353
20	threshold 0.4 (not kept)	0.354
21	ICEBERG weight 0.5, GLACIER 1.0 (not kept)	0.333
fork-v1	the public 0.421 notebook, unchanged, as our kernel	0.415
fork-v2	PubChem channel over 10,000 rows, not 5000 (not kept)	0.415
e006-fork	the public 0.446 E006 notebook, unchanged, as our kernel	0.446
e006-v3	engine-2 fusion weight 0.8, not 0.6 (not kept)	0.437
spectra-fork	the public 0.451 notebook at version 53, unchanged (not kept)	0.445
pubchemlite-fork	the public 0.456 notebook (+ PubChemLite in engine 2), unchanged	**0.456**

Two levers account for almost all of the gain: the learned spectrum model (+0.115, v5 to v6) and the forward models (+0.087, v11 to v15). Tuning the shortlist and the prior's weight (v8, v9) moved the score by under 0.01, as did removing the copied structures, although that changed 267 top-1 picks:

FPNet, trained on the copied rows, had returned their structures, and what replaced them was as rarely right. Identical runs differ by about ±0.005; v16, v17 and v18 fall inside it and were reverted. v18 is the instructive one: on 280 held-out train molecules, scored by an FPNet that never saw them, a second PubChem pass chosen by the spectrum lifted the answer's presence among the candidates from 126 to 236 and MRR@25 from 0.241 to 0.405. On the test set it moved the score by +0.004. The production FPNet was trained on the train rows the test spectra copy, and the pass's top-1 picks moved toward those copied structures (median Tanimoto 0.26 to 0.42 over the 192 molecules whose top-1 changed). v19 skips second-pass candidates with Tanimoto ≥ 0.5 to the copied structure (894 dropped over 400 molecules) and scored 0.356, +0.007 over v15. Against v18, which differs only by the filter, the gain is +0.003, inside the noise, so v19 is kept but does not clear G1; its threshold was also chosen after seeing v18. An identical re-run of v19 scored 0.353, and the 0.4 threshold (v20) 0.354: the filter's effect is not distinguishable from noise.

The fork (fork-v1) scored 0.415, +0.059 over v19 and the largest single gain after FPNet. Its commit run took 1286 s, loaded the ranker, built every base list without error, injected 600 PubChem rows in its 12 smoke molecules with no failures, and GLACIER scored 11 of the 12. Widening its PubChem channel to the top 10,000 pass-1 rows (fork-v2), chosen from our recall probe (255 against 236 of 280 held-out answers), scored 0.415 again and was discarded.

The E006 fork scored 0.446 (submission 57298936), its public score exactly, +0.031 over fork-v1 and +0.090 over v19. Its first kernel version used the source's metadata, which names Kaggle's CPU image; the notebook's T4 preflight failed (`nvidia-smi` exit 12) and it was not submitted. Version 2 runs on the GPU image fork-v1 uses. Its commit run took 1183 s: engine ready, ranker 160 features and 4 boosters, 0 base errors, ICEBERG 11 of 12 and GLACIER 11 of 11 smoke molecules scored, the engine-2 cell applied on 400 rows, and 400 output rows. Scoring the hidden test took about 4 h. Because this fork reproduced its public score on the same account and day, fork-v1's 0.006 shortfall is not a general account or scoring-day effect.

Two later submissions changed or replaced the E006 fork and neither was kept. e006-v3 changed one byte (cell 35): the engine-2 weight in the final rank fusion, 0.8 instead of 0.6. Its gate passed (ranker 160 features and 4 boosters, 0 errors, ICEBERG and GLACIER 11 smoke molecules each) and it scored 0.437 (submission 57310111), −0.009, outside the noise. dmitriigluzdov's public notebook *CASMI 26: From Spectra to Structures* reports 0.451 at its version 52 (E006 with depth-7 engine-2 rankers and standardised ChEBI/LIPID MAPS candidates); Kaggle serves only version 53, which adds ranker seeds. Forked unchanged as the private kernel `charyeezy/casmi-spectra-fork` (md5 `951a34f8...`, the author's 4 datasets, GPU image), it scored 0.445 (submission 57311145), inside the noise of 0.446.

The fourth fork (pubchemlite-fork) scored 0.456 (submission 57435777, 20 October), its public score exactly, +0.010 over the E006 fork and outside the noise. Its gate passed: ranker 160 features and 4 boosters, 0 errors, ICEBERG and GLACIER 11 smoke molecules each, and 400 output rows; the commit run took about 1190 s. Its upstream steps from the 0.451 base (PubChemLite +0.004, then N1A and popularity 0.15 +0.001) are its author's public scores; we reproduced only the endpoint.

Two offline checks, no submission. Both ran on an Apple M5 on 18 October. FRIGID, the host-approved de novo model (MIST-MSG encoder, FRIGID-base decoder), was run on 50 Enveda-180 $[M+H]^+$ molecules that have no spectra in any public library, given the true formula. The bar, fixed before the run, was the answer in the top 10 for at least 5 of 50. It found 0 of 50 (top 1, 5 and 10 alike); the best of each top 10 reached mean Tanimoto 0.230 to the answer, 3588 of 3819 generations were valid (94%, a lower bound), at 35 s per spectrum. Diagnostics run after the result locate the failure in the encoder: MIST's predicted fingerprints reach Tanimoto 0.26 to the truth on these spectra against 0.63 on MassBank, while the decoder fed the true fingerprint recovers 49 of 50 at rank 1, an upper bound inflated because its 230-million-structure pretraining probably includes these compounds. The follow-up asked whether the fold-safe FPNet could replace MIST. On 50 Enveda-180 molecules held

out of its training (re-drawn, since only 10 of the first 50 were held out), its fingerprints reach Tanimoto 0.245 in its own 10,226-bit space and 0.180 mapped to FRIGID's 4096-bit Morgan space, against 0.268 for MIST on the same molecules; it keeps only 1,464 of the 4,096 ECFP4 bits, and was trained to rank candidates rather than reconstruct bits. Retraining a 4096-bit head would take an estimated 10–15 T4 hours, not justified on this evidence.

Closed, with the reason.

- FRIGID de novo with MIST's fingerprints: 0 of 50 in the top 10 with the true formula, against a pre-registered bar of 5 of 50; the encoder (Tanimoto 0.26 on Enveda spectra) is the bottleneck.
- FRIGID driven by the fold-safe FPNet: Tanimoto 0.180 in FRIGID's bit space (0.245 in its own) against MIST's 0.268 on the same 50 held-out molecules; a 4096-bit retrain (10–15 T4 hours) not justified.
- Engine-2 fusion weight 0.8 (e006-v3): -0.009 against 0.446.
- The 0.451 notebook at version 53 (spectra-fork): 0.445, no gain over 0.446.
- PubChem promotion at $rel > 100$ on E006: not run. Decided before any submission on a reach check from public evidence: its public sibling already scored 0.444 against 0.446.
- A different GLACIER checkpoint: ours is ms-pred's public MassSpecGym-trained GLACIER; the paper's contrastive fine-tune gives 69.95 against 69.69 MassSpecGym top-1, and its NIST gain needs NIST-licensed weights, so there is nothing to swap.
- PubChem channel over 10,000 rows (fork-v2), forward-model weight 1.5 (v17), GLACIER on $[M+H]^+$ only (v16), popularity weight 0.15 (v8): inside the noise or below it.

Next. The base is now the 0.456 fork. Single settings of the E006 family look tuned (promotion threshold swept upstream, engine-2 weight 0.8 lost 0.009, forward models already at 1.0) and the de novo route failed offline, while the public lineage gained +0.035 in a week (0.421 to 0.456). The cheapest lever left is a daily check of new public notebooks, forking any credible improvement that reproduces. The leaderboard top is 0.504 (24 October). The weekly GPU quota (30 h) resets 31 October 00:00 UTC.

5 What is not established

- **Why fork-v1 scores 0.415, not 0.421.** Both our runs of the unchanged notebook score 0.006 below its public score. The E006 fork reproduced its public 0.446 exactly on the same account and day, so a general account or day effect is ruled out; the cause is not established.
- **Why the 0.451 notebook scores 0.445 for us.** Its 0.451 is reported for version 52; we could fork only version 53, which adds ranker seeds. Whether the seeds or non-reproduction explain the gap is not established.
- **Whether a de novo route could help with a better encoder.** FRIGID found nothing with MIST's fingerprints; an encoder fine-tuned on Enveda's spectra is untested. The 39–52% of the test outside PubChem and COCONUT is the SOTA sweep's estimate, not a measurement on the hidden test.
- **What in the 0.456 notebook earns its gain.** We ran it unchanged; how its +0.010 over the E006 fork divides between PubChemLite, N1A and the popularity weight is its author's public scores, not our measurement.

- **What in E006 earns its gain.** We ran the 0.446 notebook unchanged, so how its +0.031 over fork-v1 divides between engine-2 fusion, the ChEBI/LIPID MAPS candidates, the simulated ranker rows, the metric-identity overlays, the rank-2 hedge and the promotion threshold is not measured by us.
- **A per-molecule breakdown of the forks' gains.** The hidden test is not visible, so which molecules it gains on, and how many have a library match, is unknown.
- **A local estimate of the score.** Train-derived labels rated our first submission at 0.664 against 0.115 on the leaderboard. Public notebooks cannot supply labels: their downloadable outputs score only about 12 smoke-test molecules and write a placeholder (`CCO`) for the rest, since full predictions run only inside Kaggle's scoring rerun; an earlier reading of "three lineages agreeing on 12 of 400" was those placeholders, as was a local 0.030 for the 0.421 notebook. Nor can train supply a test-like held-out set: only 41 isomer pairs in 2.54 million rows share a spectrum with at least 3 peaks, and 7 of them split across the fold-safe FPNet's training as test answers and copied structures do. Each submission is the only measurement.
- **The private score.** Three choices — the shortlist size, λ, and GLACIER's input — were selected on the public split, as was v19's similarity threshold, so 0.356 is optimistic. The forks' settings were fixed by their authors before we saw any result, but tuned on the public leaderboard (the E006 notebook reports its threshold sweep there; the 0.456 notebook chose its popularity weight there), so 0.446 and 0.456 may be optimistic too.
- **Candidate recall.** Whether the answer is among the candidates cannot be separated from ranking with one MRR; a wider PubChem channel added held-out answers but no score (fork-v2).
- **Why the data is shaped as it is.** The copied train structures are the answer for at most $\sim$46 of 400 molecules.
- **The environment switch alone.** The move to a pinned Python 3.12 image and RDKit 2026.3 (v12) was not measured separately.
- **Forward-model coverage on the hidden test.** Our pipeline left 29 visible-test molecules without forward-model evidence; the forks' coverage of the hidden test is unknown.

Sources

Code and history: `https://github.com/charleneleong-ai/ai4science/pull/80`. Forked notebooks: casmi26-v465-v15-pop015 (current best), CASMI26 | E006 + PubChem Promo rel>150 | LB 0.446, CASMI 26: From Spectra to Structures and CASMI26 | v4n Fusion + PubChem on public 0.421. Every input and checkpoint with its version: `casmi/docs/experiment-card.md`. Competition: `https://www.kaggle.com/competitions/enveda-CASMI26-molecule-id-mass-spectra`.